

Justin Lee

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EDUCATION

Rice University

Houston, TX

Bachelor of Science in Computer Science, Minor in Data Science

May 2028

- **Relevant Coursework:** Object Oriented Programming, Data Structures, Algorithms, Computer Architecture.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, SQL

Technologies: React, Next.js, FastAPI, Node.js, Pandas, NumPy, Redis, LangChain, Hugging Face

Databases: PostgreSQL, MongoDB, MySQL, LanceDB

EXPERIENCE

OpenStax

Houston, TX

AI Engineer

May 2026 – Present

- Led a **5-engineer** team to architect an interactive branching-scenario application, scaling system design from schema definition to real-time production UI generating **50+** medical training modules.
- Engineered an automated book-ingestion pipeline (**BeautifulSoup**, **markdownify**) converting **5,000+** pages of unstructured textbooks into optimized Markdown, cutting data prep latency from **100+** hours to minutes.
- Designed a model-agnostic backend orchestrating interchangeable LLM endpoints (**Claude/Gemini**) and implemented rigorous prompt tuning to achieve an **80%** reduction in out-of-spec content payloads.
- Architected a **Next.js** + **FastAPI** + **PostgreSQL** stack modeling complex pedagogical pathways as a production-grade directed graph, utilizing custom **React** components to mutate AI-generated JSON nodes dynamically.

Scholarhive

Houston, TX

Founding Engineer

October 2024 – January 2025

- Scaled an edtech platform to **5.5K+** users and generated **\$500K** in verified scholarship matches; advanced company to the **YC W25** interview pool[cite: 1].
- Slashed API p95 response latencies by **85%** (from **800ms** down to **120ms**) by designing targeted SQL indexes and deploying high-performance **Redis** caching infrastructure on **AWS**[cite: 1].
- Developed an internal ML-driven matching engine achieving **94%** precision, automating complex multi-tenant data ingest across **10+** distinct sources via an optimized ETL data pipeline[cite: 1].
- Boosted user application conversion rates by **30%** by engineering a highly responsive **Next.js** discovery UI backed by efficient server-side data filtering and custom match-ranking algorithms[cite: 1].

Rice University Digital Health Initiative

Houston, TX

Software Engineer

November 2025 – March 2026

- Engineered a multi-tenant **Node.js** MVP marketplace using structured **MySQL** schemas to match **1,200+** multidisciplinary students with clinical laboratory roles via advanced multi-layered eligibility queries.
- Designed automated high-throughput data intake channels and role-based search pipelines that accelerated cross-institutional lab staffing and reduced production time-to-fill tracking metrics by **25%**.

PROJECTS

OwlConnect | *LangChain, FastAPI, MongoDB, Next.js, WebSockets, MCP*

September 2025

- Built an agentic matchmaking platform pairing academic cohorts using an embedding model and an intricate **LangChain** multi-agent negotiation system to resolve scheduling and profile alignment constraints.
- Awarded **2nd Place Overall** out of **100+** competing engineering teams at HackRice (**500+** total participants) featuring real-time, low-latency **WebSocket** visualizations mapping trajectory matching.

HootPlanner | *React, TypeScript, FastAPI, Pandas, LanceDB*

September 2024

- Awarded **1st Place Overall** out of **100+** engineering teams at HackRice (**400+** participants) for building an AI-driven, semantic course recommendation engine powered by custom **Pandas** data pipelines.
- Deployed a low-latency vector search engine utilizing Hugging Face sentence embeddings and a **LanceDB** vector database backend to achieve fast, contextual academic trajectory generation.